

Intonation in Ende¹

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QP 1

Abstract

This paper employs a contour clustering analysis (Kaland 2023) to explore whether or not intonation plays a role in distinguishing between declarative, interrogative and imperative sentences in Ende. The results of both the cluster analysis and a qualitative evaluation of Ende phrases suggest that intonation does not distinguish between these sentence types in Ende. However, this evaluation does indicate that the f0 contours over Ende phrases share a consistent shape characterized by a pitch peak at the beginning of the phrase and a fall to the pitch minimum at the end of the phrase. A discussion of the patterns exhibited by these pitch movements suggest that Ende's intonational system plays a role in delineating the boundaries of prosodic units.

1 Introduction

This paper presents a preliminary investigation of the intonational system of Ende (ISO639-3 code: kit), an underdocumented language of Papua New Guinea. This study has a few basic goals. The overarching goal of this paper is to address the efficacy of a contour cluster analysis as an exploratory tool in the early stages of intonational investigations using a corpus of narratives. Within this, the primary goal is to use this tool in order to address claims that intonation is not used to distinguish between *declarative* and *interrogative* phrases in Ende (Lindsey 2022). The secondary goal, is then to explore additional insights into Ende's intonational system that the cluster analysis may reveal. While the cluster analysis suggests that intonation in Ende does not play a direct role in distinguishing between sentence types, it does identify a typical contour shape which is characterized by a rise in pitch at the beginning of the phrase and a fall in pitch to the end of the phrase. A preliminary inspection of the distribution of these pitch movements provides potential support for three prosodic units which align roughly with a *word*, a full *phrase* or *utterance*, and something that falls in between. As such, the cluster analysis appears to be a useful tool at the beginning stages of intonational description, and especially in contexts where the researcher is constrained to the use of previously collected corpora. This tool not only provides support for previous claims concerning the relationship between intonation and sentence type in Ende, but it also revealed consistent pattern in the intonational contours of Ende phrases which hint at prosodic constituency and offer promising directions for future investigations.

This presentation begins with a brief overview of the cross-linguistic study of intonational systems. This is followed by a discussion of the under-representation of intonational descriptions

¹ I am grateful to Professors Kate L. Lindsey, Jon Barnes and Danny Erker for their guidance, patience and support. I would like to acknowledge the contributions of the members of the Ende tribe whose voices and knowledge made this work possible. Finally, I would like to thank the audience at the 15th annual conference for *Austronesian and Papuan Languages and Linguistics* for the discussion and ideas which will guide the future directions of this work. All errors are my own.

across the world's languages and some emerging methodologies aimed at remedying the situation. The intention of this discussion is to provide a backdrop against which the following investigation of Ende's intonation may be contextualized, as well as to offer a motivation for this exploratory investigation into Ende's intonational system.

The study of prosody deals largely with suprasegmental features of speech (e.g. duration, intensity, and pitch). Within this field, *intonation* is generally used to refer to the systematic differences in pitch at the phrase, utterance, or clausal level (Cruttenden 1997, Ladd 2008, Zerbian 2010). In many languages, differences in pitch both between and within utterances may be used to communicate meaning across a number of domains. Intonation may be used to communicate social information pertaining to a speaker's stance or affect. For example, a high or raised pitch may be associated with a speaker's lack of confidence or deference to the interlocutor while a low or falling pitch may be associated with a speaker's confidence, aggression, or assertion of dominance in relation to their interlocutor (Bolinger 1978, Ohala 1984, 1996). At the same time, these differences in pitch may be used to distinguish between sentence types. Again, the literature reports a general divide between a high or rising pitch to signal interrogative sentences with a low or falling pitch to signal declarative sentences. This divide may be exemplified with an English sentence such as *He went to the store*, which is taken to be a question when uttered with a rise in pitch at the end but is taken as a statement of fact when uttered with a falling pitch. Beyond this movements in pitch may be used to signal the boundaries of prosodic units (a demarcative function) or they may be used to mark various aspects of information structure, such as focus or topic (a highlighting function).

Intonation often plays many of these roles at the same time and as such, constitutes a rich area for linguistic inquiry. Despite this, information on intonation is often omitted from phonetic and phonological descriptions. A survey performed by Whalen, DiCanio, and Dockum (2022) illustrates the severity of this gap in the literature, reporting that only 16% of the phonetic illustrations published in the Journal of the International Phonetic Association, 8% of articles in the Journal of Phonetics, and 10% of entries in Sounds of the World's languages address intonation. This gap stems partially from the complexity involved in untangling intonation's use in these functions from one another, and also from a number of practical issues surrounding language description in general. First, phonetic and phonological description depends on the ability to make and analyze high quality sound files, two tasks which are difficult without the availability of specialized equipment and software. Fortunately, advances in technology have made audio-visual recording devices more accessible and portable, coupled with the development of free software for acoustic analysis such as Praat (Boersma & Weenink 2018) have made it easier to collect and analyze phonetic data in the field. Additionally, the development of methodological guides for fieldworkers (e.g. Himmelmann 2006, Himmelmann & Ladd 2008, Jun & Fletcher 2014) have made it easier for researchers to navigate the literature surrounding speech prosody and employ the specialized elicitation methods for prosodic and intonational descriptions.

Although these improvements make the task of describing a language's intonational system more approachable, there are still a number of obstacles remaining. Principle among these is the

difficulty one might face is in getting access to speakers to participate in the necessary experiments and elicitations. In many cases, the only way to do this is to travel to the communities in which the language in question is spoken. However, these are often in remote locations that are difficult to get to. In the best of times travel to these areas requires the commitment of a considerable amount of time and resources on the researcher's part. Similarly, supporting the fieldworker can be draining on the community. Since the advent of COVID-19, these issues have only been compounded by the imposition of travel restrictions and the increased risk to the health of these communities. As such, it is well worth developing and testing creative methods to continue documentation efforts without boots-on-the-ground field work. One path towards this goal is through the use of audio data housed in publicly accessible online corpora.²

There is already a small body of work, assessing the potential use of publicly accessible corpora for phonetic investigations in understudied languages. Babinski et al. (2022) provide an assessment of the extent to which a sample of digital corpora are amenable to phonetic analysis and offer some recommendations for improvement in this area. Furthermore, Babinski (2022) demonstrates the efficacy of using these corpora along with quantitative methods to investigate the word-level prosodic structures in understudied (and even dormant) languages of Australia. Looking at phenomena above the segmental and word-level, Siefert (2021) presents two case studies investigating speech rate and introducing a time aligned corpus of 50 languages intended for use in phonetic investigations. Looking specifically at intonation, Kaland (2023) introduced a workflow for applying a cluster analysis to intonational contours which he argues may be applied to corpora of spontaneous speech in order to identify potentially meaningful movements in f0 at the early stages of the description of an under-documented language's intonational systems. Babinski and Bower (2022) and Babinski (2022) present case studies in which this cluster analysis workflow is applied to archived corpora.

This paper contributes to the field of intonational research by presenting an application of Kaland's (2023) cluster analysis workflow to develop a preliminary hypothesis about the intonational system of Ende, an underdocumented language of Papua New Guinea. This study utilizes publicly available resources housed in the *Language Corpus of Ende and Other Pahoturi River Languages* (Lindsey 2015). As discussed in the opening paragraph, the cluster analysis corroborates the previous characterization of intonation in declarative and interrogative phrases presented by Lindsey (2022). Additionally, this contour cluster workflow helps to elucidate a number of pitch movements which appear consistently across Ende phrases. A preliminary analysis of the distribution of these pitch movements helps develop a preliminary understanding of intonation's role in demarcating prosodic units in Ende speech.

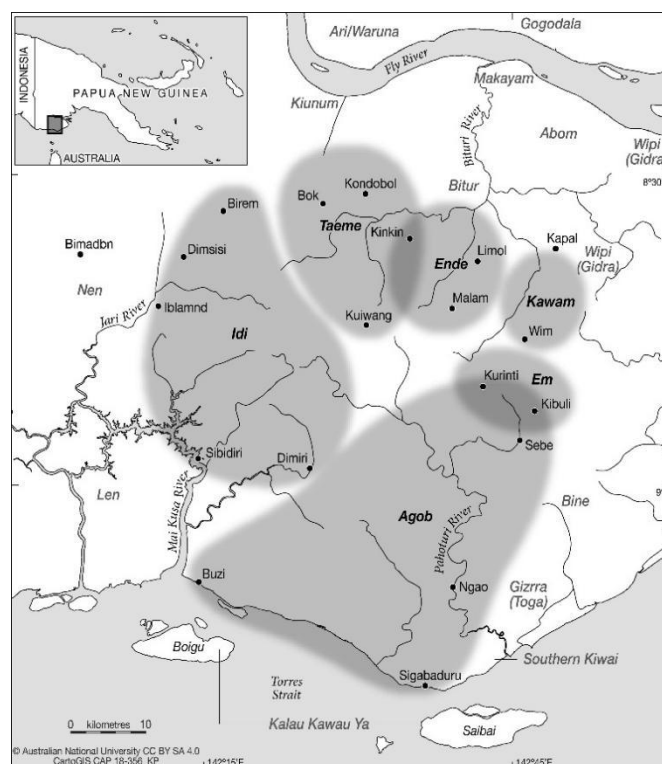
Section 2 provides background concerning the language Ende, the area in which it is spoken, and the current state of intonational description among the languages of Southern New Guinea. Section 3 discusses the methodology employed in this investigation, namely a contour clustering

² It should be noted that in many cases the primary goal of creating these corpora is to preserve the various facets of cultural knowledge that may be lost to language dormancy and/or serve in the process of language revitalization (Austin 2021, Fitzgerald 2021) and not necessarily to act as data for linguistic studies. In fact, even when the focus of a corpus is linguistic in nature it is very seldom done so with the intention of investigating intonation.

analysis (Kaland 2023). Section 4 reports on the findings of the cluster analysis and discusses the relationship between sentence type and intonation in Ende. The cluster analysis suggests that the typical Ende phrase, be it *interrogative*, *declarative*, or *imperative* is characterized by an intonational contour with a pitch peak near the beginning of the phrase, a pitch fall to the end of the phrase. This generalization is then shown to hold in the phrases present in two independent sound files which were not included in the dataset for the cluster analysis. Section 5 briefly discusses the implications of these findings in relation to what little is known about the intonational systems of other languages in the area of Southern New Guinea. Then, Section 5 explores the distribution of these two pitch movements and discusses how the positions of these pitch movements may signal the boundaries of different prosodic units such as the word, the phrase, and something in between. Section 6 offers some concluding remarks.

2 Background

This paper addresses issues surrounding the documentation of phrase-level prosody in Ende. This study employs an intonational contour based cluster analysis (Kaland 2023) to address the claim that intonation does not play a role in distinguishing between declaratives and interrogatives in Ende (Lindsey 2022). It is estimated that there are between 600 and 1000 speakers of Ende primarily spoken in the villages of Limol, Malam, and Kinkin (see Map 1; Lindsey 2022). Ende is a member of the Pahoturi River language family spoken in Western Province, of southern Papua New Guinea. Along with Ende, the Pahoturi River family is comprised of Agob, Idi, Taame, Kawam, Em and Idan (Lindsey, Schokkin, & Wu 2022, Scanlon 2023).



Map 1: The geographic primarily Pahoturi River language speaking villages (Lindsey 2019). The villages and geographic area in which Idan is primarily spoken is not represented on this map. This is because Idan's existence was only made known to linguists working in the Pahoturi River after this map's creation.

2.1 Intonational descriptions in southern New Guinea

Currently, Ende is the only Pahoturi River language with published information pertaining to its intonational system. This is detailed in Section 2.2 below. The Pahoturi River languages are situated within Southern New Guinea, an area well known for its high linguistic diversity (Evans, et al. 2018). Many of the known languages spoken in this area are drastically underdocumented. Among those languages for which documentation is available, intonation is rarely touched on per Whalen et al. (2022). This is exemplified in the special edition of the journal *Documentation and Conservation Phonetic Fieldwork in Southern New Guinea* (Lindsey and Schokkin 2021). Out of the seven languages included in the issue, only two – Yelmek (Gregor 2021b) and Nmbo (Kashima 2021) – contain descriptions of their intonational systems.³ Beyond this special edition, a number of languages spoken in Southern New Guinea have published grammars which offer brief insights into their intonational systems.

In both Yelmek (Gregor 2021a, b) and Nmbo (Kashima 2021), intonation does not appear to play a clear role in distinguishing between declarative and interrogative phrases. In both of these languages, both declarative and interrogative phrases have intonational contours which decline over the course of the utterance.

In Yelmek both declarative and interrogative phrases are described as having falling intonation (Gregor 2021a, b). Additionally, the description presented by Gregor (2021a) suggests that phrases in Yelmek tend to have a pitch peak occurring over the final syllable of the first word in the phrase. However, in cases where an element of the phrase is under focus, the pitch peak is realized over the focused word, which bears a focus suffix *-i* and typically surfaces as the second word in the phrase.

Phrases in Nmbo tend to have relatively level pitch until a clear fall at the end of the phrase (Kashima 2021). Kashima (2021 pp. 66 - 67) describes three non-falling final boundary markers in Nmbo: a “flat final boundary” which may signal a shift in discourse topic like the typical falling boundary marker, a continuation boundary marker which appears to be realized similarly to the “flat final boundary”, and a “rising final boundary” which apparently functions similarly to the falling final boundary in marking the end of a clause and intonation unit.

Beyond the special issue, I have been able to find six grammars for languages spoken in Southern New Guinea: *A grammar of Komnzo* (Döhler 2018), *A sketch grammar of Meryam Mir* (Piper 1989), *A documentation and description of Yelmek* (Gregor 2021a), *A grammar of Suki* (van Tongeren 2022), *A Grammar of Nama: A Papuan Language of Southern New Guinea* (Siegel 2023) and *The Coastal Marind language* (Olsson 2018). With the exception of Yelmek (Gregor 2021a), these grammars do not offer dedicated descriptions of either the phonetics or the phonology of intonation in their respective languages. Despite this there is some information pertaining to intonation included. Each grammar references an “intonation group”. This term is typically undefined but seems to suggest that intonation plays a role in organizing the speech

³ 28% of the descriptions in this issue address intonation. This is a much higher rate than is seen for the resources surveyed in Whalen et al (2022). However, this rate is likely inflated by the small sample size included in the issue.

stream into discrete units (i.e. a demarcative role). Additionally, Komnzo, Meryam Mir and Nama are described as having a rising intonation which distinguishes polar questions from declaratives. Coastal Marind is described as having rising intonational contours which signal continuation and a falling contour which signal completion when a sequence of clauses are linked together.

Although the articles presented in the special issue and the descriptions in the grammar offer some insights into the intonational systems of different languages spoken throughout Southern New Guinea, only Yelmek and Nmbo offer phonetic details beyond the terms of “rising” and “falling” intonation.

2.2 Prosody in Ende

At the word-level, Ende does not appear to have either lexically contrastive tone or stress (Lindsey 2022). Lindsey (ibid) describes the use of intonation in a “Call-at-Distance form” in which the addition of /-o:/ or /-aj/ is added to the end of an utterance along with a considerable rise in pitch and loudness. This raised pitch may occur over the entirety of the phrases which often terminates in a steep fall.

Outside of this specific construction, however, Ende’s intonational system remains largely undescribed. Lindsey (2022) states that interrogative phrases do not systematically end in a rising pitch and are often characterized by a falling tune, much like Ende’s declarative phrases. However, this characterization is based largely on the researcher’s impressions and is not accompanied by a detailed phonetic study. The investigation presented in this paper offers some more detailed support for this characterization of intonation in Ende phrases.

2.3 Intonational research using corpora: contour clustering analysis

The remoteness of the communities in which the languages of Southern New Guinea are spoken has presented a major obstacle for efforts to document them. This Additionally, even if researchers are able to circumvent these obstacles, other factors, such as civil or political unrest may disrupt their plans⁴.

In cases when working with native speakers is not feasible, accessible corpora present opportunities to do research beyond the scope of initial fieldwork. However, the existence of an accessible corpus does not guarantee that the kinds of elicitation paradigms typically used to investigate intonation are present or identifiable within the corpus. Kaland (2023), presents a toolkit and workflow for the application of a cluster analysis intended to help researchers develop and test hypothesis in the early stages of intonational research. This methodology employs an automated bottom up workflow which Kaland argues can help to identify subtle differences in intonational contours that may be missed by a non-native researcher while circumventing the biases introduced by intonational categories in the researcher’s native language. This workflow groups intonational contours according to phonetic similarity using a complete-linkage clustering algorithm (discussed further in section 3.1). It is then the job of the

⁴ This, in fact, is the situation which led to the formulation of my investigation: a trip that was planned to go to the area in the summer of 2022 had to be cancelled due to election related violence in the Port Moresby and Daru, after having been plagued by visa related issues.

researcher to determine whether these groups are coherent and how the forms of the identified contours map onto their functions.

This toolkit has already been used in a number of different studies that may help to elucidate its usefulness. Seeliger and Kaland (2022) present an application of this method to German phrases containing *wh*-words. They argue that the clustering algorithm is able to successfully categorize the *wh*-exclamative from *wh*-questions which differ primarily in the shape of pitch movements at the end of the phrase. There are also a number of cases where this workflow is applied to corpora of spontaneous speech. Kaland (2023, sections 3) applies the cluster analysis workflow to a corpus of pear story narrations in Papuan Malay; Babinski and Bower (2022) apply it to a corpus of Bardi narratives; and Babinski (2022) applies it to narrative corpora for 16 indigenous languages of Australia. These applications do not explicitly identify intonational distinctions between interrogative and non-interrogative phrases. Babinski and Bower attribute this to the low number of interrogative phrases that are represented in narratives. However, these case studies do demonstrate the range of phonetic differences that the algorithm is likely to group contours according to. For instance, Kaland's presentation of Papuan Malay Pear Stories identifies a primary split between a large cluster ($n = 262$) with low flat averaged contour and a much smaller cluster ($n = 44$) which has an averaged contour that shows a rise and plateau shape beginning at the middle of the contour. Subsequent splits reveals that the initial large cluster with a flat averaged contour could be divided into smaller clusters with contours that differ in overall pitch height, the alignment of pitch peaks throughout the phrase, and the trajectory of pitch movements near the end of the phrase (i.e. rising vs. falling contours).

3 Methodology

The overarching goal of this paper is to explore the efficacy of Kaland's (2023) contour clustering toolkit and workflow in the early stages of investigating the intonational system of Ende. Within this there are two more specific goals. The first is to address the claim that Ende does not use intonation to distinguish between sentence types like declaratives and interrogatives (Lindsey 2022), and in so doing gain a clearer picture of the relationship between phonetic form and sentence type in Ende. Although there does not appear to be a clear relationship between intonation and sentence type in Ende phrases, the cluster analysis seems to identify a relatively homogeneous contour shape which is characterized by a rise to the pitch maximum near the beginning of the phrase and a fall to the pitch minimum at the end of the phrase. As such, the second goal of this paper is first to confirm that the shape identified by the cluster analysis does appear to be consistent across phrases in Ende and then to explore the distribution of pitch movements associated with the boundaries of the Ende phrase and attempt to relate them to the function of delineating prosodic units of different sizes. The remainder of this section outlines the data and methodology employed within this study. In the discussion throughout this section, capitalization will be used to indicate a variable name or value.

3.1 The Language Corpus of Ende

3.1.1 Files used for cluster analysis

The files that are used in this study are all publicly available through *The Language Corpus of Ende and other Pahoturi River Languages* (Lindsey 2015) a publicly accessible corpus hosted on PARADISEC. The corpus contains audio-visual recordings of word lists, narratives, sociolinguistics interviews and a host of other elicitation tasks. A large portion of the files contained in this corpus have been segmented, transcribed, and translated.

For the cluster analysis, eight files were selected from *The Language Corpus of Ende* (Lindsey 2015). These files were contributed to the corpus by two male speakers: Tony Warama and Jerry Dareda. These eight files contain four different stories, each of which was told and recorded twice. The original version of the stories were written by the speaker before the recording session and then read aloud. Each story was then retold from memory by the same speaker. *How Cuscus' ears got short* (Warama 2016a, b) is about two friends Dog and Cuscus who love to play hide and seek. During one fateful game of hide and seek Dog tricks Cuscus into cutting his ears off. As a result, Cuscus curses Dog, so that he will always run around on all fours eating rubbish from humans. In return, Dog curses Cuscus so he will spend his life eating tree leaves being forever hunted by humans. In *Island hunting*, Tony Warama (2016c, d) tells the personal story about hunting with his friends on Dikullwang island. *Bundae's story* (Dareda 2016 b, c) tells the story of young boys who are hunting and come across a man named Bundae who had a very big head. Bundae was sleeping, so the boys decided to camp inside his nostril for the night. In the end, Bundae sneezes the boys out of his nostril and teaches them a lesson about respecting other people's space. Finally, in the *Watermelon story* Jerry Dareda (2016a, d) tells a personal story about going to Upiara with a friend to harvest watermelons.

3.1.2 Files used for turning point analysis

The cluster analysis is followed by a follow up inspection of two additional sound files. The first file is a dictionary word elicitation by Tony Warama (Warama 2017) and the second file is a Functions of Language elicitation with Linda and Anna Duiya (Duiya & Duiya 2018). In the dictionary words elicitation T. Warama presents a list of words for an Ende dictionary. In this file he reads each word, the definition, and an example sentence in Ende. For this analysis, only the example sentences were considered. The words in isolation were not considered here because they present cases in which the word and phrase level phenomena may be conflated. The definitions were not considered because they tend to be noun phrases.

The Functions of Language elicitation (Duiya & Duiya 2018) is intended to capture language use across a number of everyday contexts that aren't likely to arise during typical linguistic elicitation. In each case, the fieldworker presents the Ende speakers with a situation and asks them to act it out. Although this elicitation paradigm does not perfectly replicate naturalistic international language use, it is the closest to doing so among the files included in this study.

3.2 Segmentation and annotation

The files used in this study were originally segmented, transcribed, and translated by Dr. Kate L. Lindsey in collaboration with Ende speakers. The files are all available as ELAN files in *The Language Corpus of Ende* (Lindsey 2015). The Praat TextGrids (Boersma & Weenink 2018) that were exported from these files were adapted by the author of this paper for this study. The annotated and adapted TextGrid files are currently not publicly available.

In order to avoid any potential biases concerning phrasehood that could be introduced by the author's native language, boundaries were initially set at the beginnings and endings of pauses longer than 10 ms in duration. Since the primary goal of this investigation is to address the claim that Ende does not use intonation to distinguish between sentence types (e.g. declarative vs. interrogative), the segmentation was intended to target large prosodic units approximately the

size of a clause. Since Ende's word order is largely verb final, boundaries that did not coincide with a verb were removed. However, there is high degree of word order variation which might strand elements of a phrase to the right of the verb (Brown 2020); when these cases could be identified, the stranded element was included in the interval containing the verb to which it was related. Additionally, phrases with sentence type other than *declarative* only appeared in quotations. For this reason, quotations representing the speech of the characters were segmented into their own intervals, excluding the portion of the narrative within which they were embedded. In this process, false starts and speech errors were excluded whenever possible.

Each interval was annotated according to two features: (1) whether the phrase was part of the Narrative (Narr) or a Quotation (Quo) from one of the characters, and (2) the sentence type that is represented by the phrase. The sentence types that were coded for were Declarative (DEC), Content Question (CQ), Polar Question (PQ), Imperative (IMP), Call (CALL), Exclamation (EXC), and Answer (ANS). The combination of the two annotation schemes is such that a Declarative phrase in a narrative context is labeled as Narr-DEC while a Declarative phrase found in a quotation is labeled as Quo-Dec. During post-processing (the procedures for which are outlined in Section 3.4 below), these annotations are used to assign each phrase the appropriate values for a variable indicating its sentence type and whether or not it was uttered in a narrative context. Declaratives were defined as any phrase offering a statement or description. Content questions (or Wh-questions) were defined as any phrase asking a question which contained a word like *what*, *where*, *when*, *why*, or *how*. Polar (or yes-no) questions were defined as any question which could be answered with a *yes* or a *no* and are distinguished from content questions in that they do not contain question words. Imperatives were defined as any phrase in which the speaker is issuing a command. Calls were defined as any phrase in which a character is calling the attention of another character. Exclamations were defined as any phrase which expressed a reaction like *oh wow* or *achoo!* Answers are defined as any phrase in which a character is directly answering a question with the words *Ao* 'yes', *Ause* 'yes', *ddone* 'no' or *ka* 'no' alone.

The segmentation guidelines used for the dictionary elicitation (Warama 2017) and the Functions of Language elicitation (Duiya & Duiya 2018) are identical to those used for the narrative files. Each phrase was annotated according to its Sentence Type. The speech in these files was not narrative in nature and as such the distinction between Narrative and Quotation could not be coherently maintained for the phrases in these files.

3.3 Data Subsets for cluster analysis

A few different data subsets are used throughout the presentation of the cluster analysis data. The cluster analysis was run on the entire set of phrases segmented from the eight narratives discussed above in Section 3.1.1. As such, the visualization showing the averaged contour for each identified cluster reflects the averages as calculated over the entire dataset. On the other hand, the visualizations and conditional inference trees exploring the relationships between the identified clusters and the sentence types of interest only include the phrases that were embedded in quotations and that had a Sentence Type value of Content Question, Declarative, Imperative, or Polar Question.

This subset was selected for the visualizations and the conditional inference trees in order to make the patterns more apparent. Narrative phrases were excluded because they are exclusively Declarative. This leads to an asymmetry within the data that obscures potential relationships between Sentence Type and Cluster. As Babinski and Bower (2023) suggest, this asymmetry may also make it harder for the clustering algorithm to identify a distinct cluster for interrogative phrases to the exclusion of other Sentence Types. Phrases with Sentence Type values of Answer, Call, and Exclamation were excluded because these Sentence Types fall outside the scope of the primary goals of this investigation.

3.4 Cluster Analysis

This analysis uses the contour cluster toolkit developed by Kaland (2023)⁵ to identify groups of phrases which may share intonational contours with similar phonetic shapes. This toolkit includes two scripts: a Praat script which extracts raw f0 measurements at 20 time-steps per interval and an R script which launches a Graphical User Interface (GUI) which is used to run cluster analysis. F0 measurements are extracted using the “time series f0” Praat script. This was done with minimal modifications to the script. The script takes 20 measurements across all intervals containing text on the specified TextGrid tier (in this case, the tier for sentence type). The default settings from Kaland (2023) were kept the same. The pitch range is set from 75 to 600 Hz; time-step is set to 0.01; and stylization is set to 2 st.

The cluster analysis is run using the contour clustering GUI (Kaland 2023) launched through RStudio (R Studio Team 2019). This employs an agglomerative hierarchical clustering algorithm with complete linkage clustering in order to group f0 contours into clusters based on measures of maximum distance. The analysis proceeds through an iterative bottom-up process in which clusters are successively merged based on calculations of least maximum distance between clusters. In other words, each intonation contour is treated as its own independent cluster then compared and merged with the next closest cluster. This process is then repeated until all sub-groupings of contour clusters are sorted into two maximally different clusters.

In order to control for differences between the two speakers’ pitch range, the data was standardized by dividing the difference of each f0 measurement and the mean f0 of the file that it was extracted from by that file’s f0 standard deviation⁶.

The investigation of potentially meaningful clusters followed the procedure demonstrated by Kaland (2023). First, the data was pruned with the intention of excluding potentially anomalous clusters containing f0 measurement errors. This was done by setting the number of clusters $n=25$ and excluding clusters according to the automatic sub-setting suggestions provided by the GUI. Then, the number of clusters was set to $n=2$ and increased by a size of $n+1$ until splitting the dataset into more clusters seemed uninformative. This was in part done by judging whether the additional split provided a cluster whose shape was noticeably different from those already presented, in part by judging whether the split produced a noticeable change in the shape of any of the existing clusters, and in part by inspecting whether the split provided any informative change in the distribution of sentence type across the clusters. Since Kaland (2023), the GUI has also been updated to provide an information cost analysis which can help the researcher settle on the number of clusters most appropriate for the given dataset.

⁵ The manual, tutorial, and all requisite scripts for the toolkit described in Kaland 2021 are available at the following link: [Contour clustering \(constantijnkaland.github.io\)](https://constantijnkaland.github.io).

⁶ Kaland (2021)’s script uses filename as a proxy for speaker.

4 Analysis

4.1 Cluster analysis

620 intonational contours were input into Kaland's (2023) GUI for the initial cluster analysis. 25 clusters were generated and clusters were pruned from the dataset following the automatic recommendations made by the GUI. After three rounds of subsetting, none of the remaining clusters were flagged for pruning. This left a total of 525 clusters (84.7% of the original dataset).

Following the procedure outlined by Kaland (2023) a range of clustering solutions starting with 2 and increasing by 1 were inspected to determine the most appropriate number of splits for the dataset and the question that we wished to answer. Since the primary question in this case concerns the relationship between sentence type and intonation we also inspected the distribution of sentence types over the identified clusters and vice versa. This procedure indicates that a split of two clusters is the most informative for understanding the relationship between intonation and sentence type in Ende. The discussion below details the results of this analysis. The use of capitalization in the following discussion is intended to indicate the direct reference of a variable or the value of a variable. For example, the word 'Declarative' should be understood to refer to a phrases which were coded as declarative in the dataset.

Figure 1 below shows the averaged contours and the size of the clusters identified for a two cluster solution.

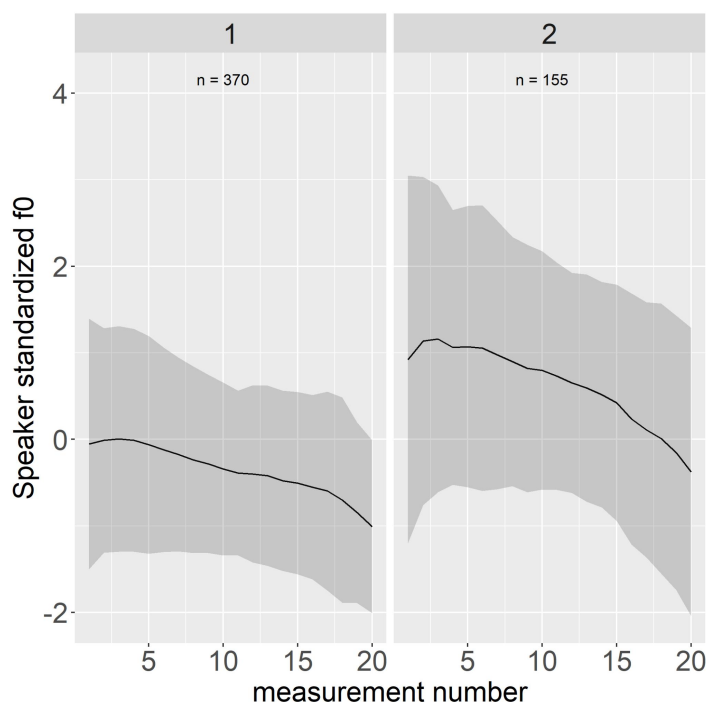


Figure 1: The averaged contour across 20 f0 measurements for each of the two clusters identified. Cluster 1 is composed of 351 phrases while Cluster 2 is composed of 158 phrases. Cluster 1's averaged contour occupies a lower pitch range when compared to Cluster 2.

The averaged contours for the two identified clusters differ in a number of ways. First, Cluster 1 appears to occupy a low and more narrow space in the standardized pitch range than

Cluster 2 does. Additionally, Cluster 2 has a more pronounced pitch peak near the beginning of the phrase and a steeper global fall to the end of the phrase. Despite these differences, both averaged contours share similar shapes. Both appear to have a globally falling shape which are characterized by a rise to the pitch maximum near the beginning of the phrase and a fall to the pitch minimum at the end of the phrase. It should also be noted that the error space around both averaged contours indicates quite a bit of overlap in phonetic form between the contours sorted into Cluster 1 and those sorted into Cluster 2. This may suggest that the two averaged contours are not indicative of two distinct intonational categories, but are instead two points along a phonetic spectrum.

The results of the cluster analysis will be presented and discussed from two different perspectives. The first presentation will show the distribution of Sentence Type over Cluster and will discuss this as a task of predicting the Sentence Type from Cluster. Figure 2 shows the distribution of Sentence Types over the two Clusters while Table 1 reports the exact number of each Sentence Type in each Cluster. Figure 3 shows a conditional inference tree with Sentence Type as the dependent variable and Cluster, Speaker, Duration (of the phrase in seconds), and whether the phrase was from a Read or Retold narrative as the independent variables.

The second presentation will show the distribution of Cluster over Sentence Type and will discuss this as a task of predicting Cluster by Sentence Type. Figure 4 shows the proportional distribution of Cluster over Sentence Type. Table 2 shows the reports the exact numbers of each Cluster in each Sentence Type. Figure 5 shows the output of a conditional inference with Cluster as the dependent variable and with Sentence Type, Speaker, Duration of the phrase (in seconds) and Read vs. Retelling as the independent variables.

This section concludes with a brief discussion of the 5 cluster split.

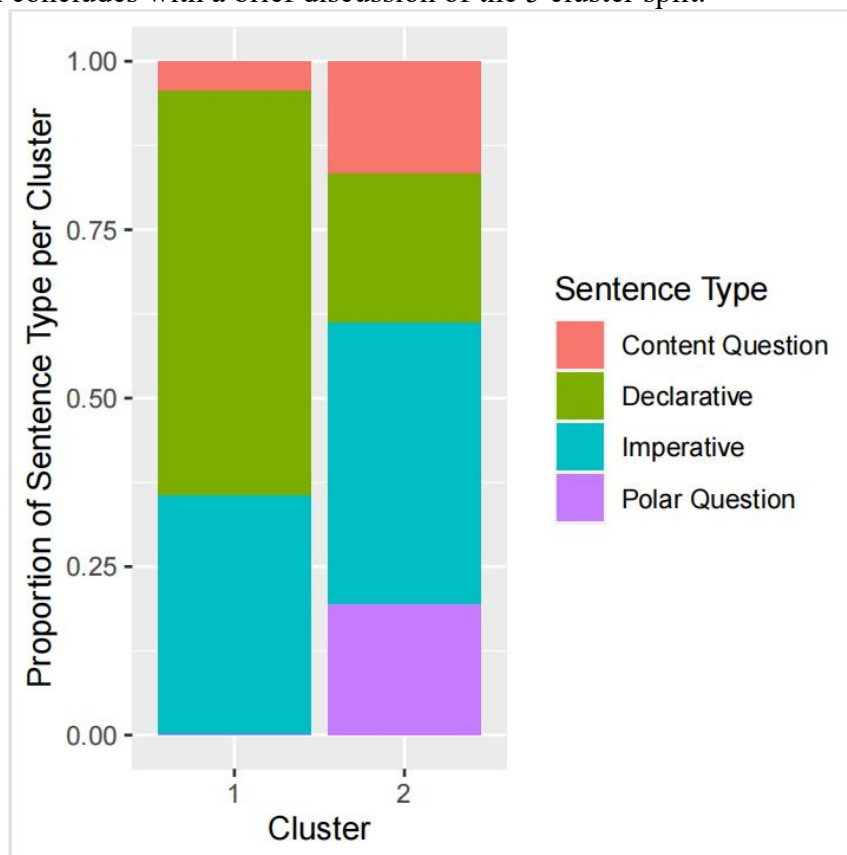


Figure 2: Sentence type (color coded) across cluster.

The most apparent trend in the distribution of Sentence Type over Cluster is that Cluster 1 is almost entirely composed of phrases with a Declarative or Imperative Sentence Type. There are 0 Polar Questions and only 2 Content Questions meaning that phrases asking a question of any kind only make up 4.4% of the total phrases assigned to Cluster 1. Cluster 2 has a more even distribution of Sentence Types when compared to Cluster 1. 23% of the contours sorted into Cluster 2 are from phrases with a Declarative Sentence Type; 42% are from Imperative phrases; 16% are from Content Questions; and 19% are from Polar Questions.

	Cluster 1	Cluster 2	Total
Declarative	27	8	35
Imperative	16	15	31
Content Question	2	6	8
Polar Question	0	7	7
	45	36	

Table 1: Numerical distribution of Sentence Type over Cluster. Of the 35 Declarative phrases uttered in quotations 27 were assigned to Cluster 1 and 8 were assigned to Cluster 2. Of the 31 Imperative phrases uttered in quotations, 16 were assigned to Cluster 1 and 15 were assigned to Cluster 2. Of the 8 Content Questions uttered in quotations, 2 were assigned to Cluster 1 and 6 were assigned to Cluster 2. Of the 7 Polar Questions uttered in quotations, 0 were assigned to Cluster 1 and 7 were assigned to Cluster 2.

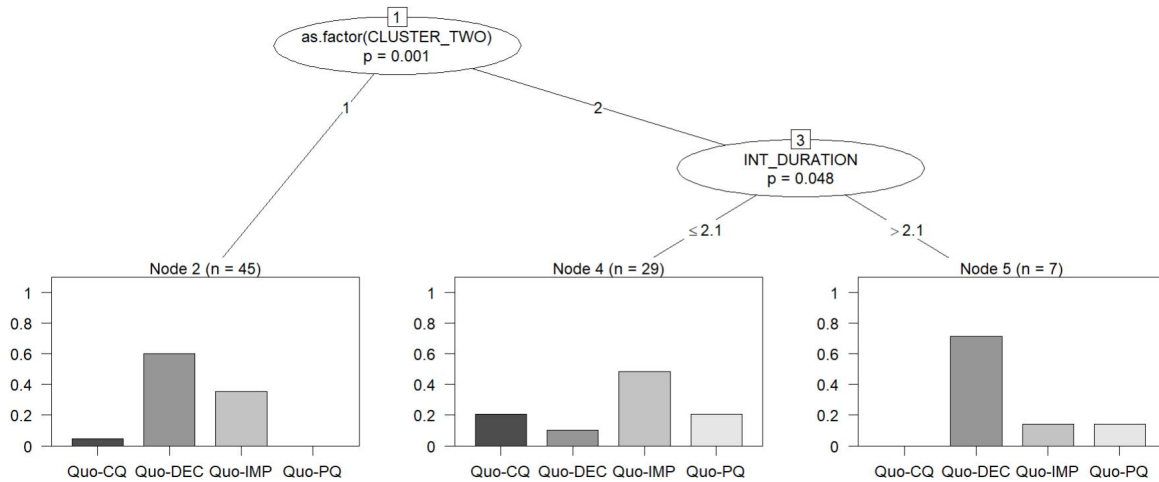


Figure 3: Conditional inference tree with Sentence Type as the dependent variable and with Cluster, Speaker, Phrase duration (in seconds) and Read vs. Retelling as independent variables. Here the variable tag Quo-CQ corresponds to the Sentence Type Content Question, the tag Quo-DEC corresponds to the Sentence Type Declarative, Quo-IMP corresponds to the Sentence Type Imperative and Quo-PQ corresponds to the Sentence Type Polar Question. The first split is between phrases sorted into Cluster 1 vs. Phrases sorted into Cluster 2. Within Cluster 2 there is a secondary split between phrases shorter than or equal to 2.1 seconds in duration and those longer than 2.1 seconds in duration.

Figure 3 shows the output of a conditional inference tree with Sentence Type as the dependent variable and with Cluster, Speaker, Duration of the phrase (in seconds) and Read vs.

Retelling as independent variables. The first split identified in the tree is between phrases assigned to Cluster 1 and phrases assigned to Cluster 2 ($p = .001$). The proportional bar graph for the set identified as Cluster 1 shows a similar distribution of Sentence Types as that shown in Figure 2. These phrases are about 60% Declarative (Quo-DEC) and about 40% Imperative (Quo-IMP), with the Content Questions (Quo-CQ) making up less than 10% of the Cluster. Within Cluster 2, there is a second split between phrases that are shorter than or equal to 2.1 seconds in duration and those that are longer than 2.1 seconds in duration. This split isolates most of the remaining Declarative phrases from the rest of Cluster 2, with phrases in Cluster 2 being much more likely to be Declarative if they are longer than 2.1 seconds. This specific relationship between Sentence Type, Cluster, and Duration may be a consequence of the f0 measurement procedure used in this workflow, the implications of which will be discussed further in Section 4.1.1.

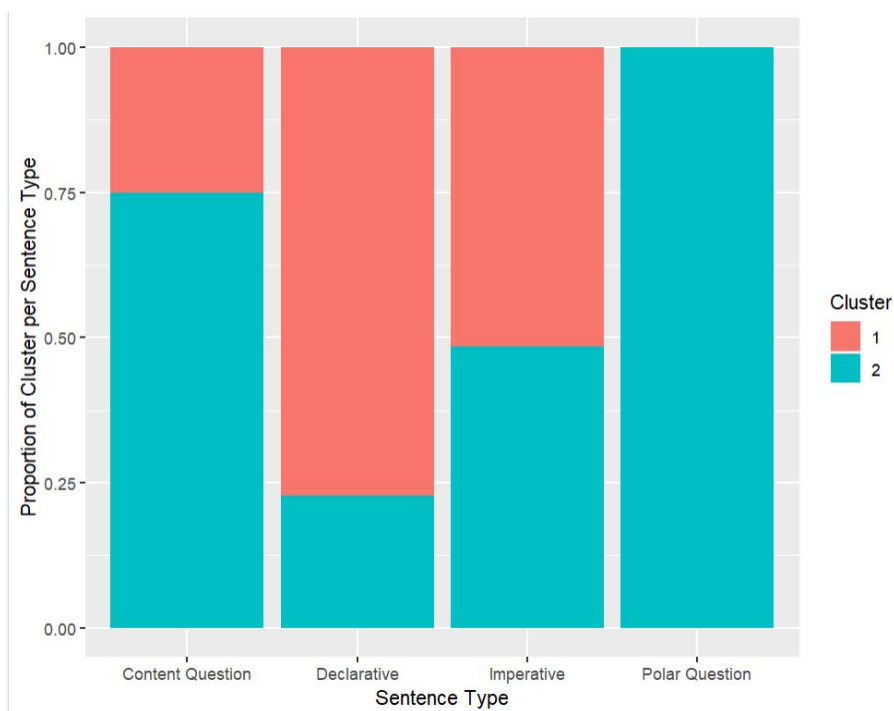


Figure 4: Distribution of Clusters over Phrase Type. Of the phrases with a Content Question Sentence Type, 25% of their contours are sorted into Cluster 1 and 75% of their contours are sorted into Cluster 2. Of the phrases with a Declarative Sentence Type, 71% of their contours are sorted into Cluster 1 and 29% of their contours are sorted into Cluster 2. Of the phrases with a Imperative Sentence Type, 52% of their contours are sorted into Cluster 1 and 48% of their contours are sorted into Cluster 2. Of the phrases with a Polar Question Sentence Type, 0% of their contours are sorted into Cluster 1 and 100% of their contours are sorted into Cluster 2.

	Declarative	Imperative	Content Question	Polar Question
Cluster 1	27	16	2	0
Cluster 2	8	15	6	7

Table 2: Numerical distribution for Cluster over Sentence Type. 27 of the phrases assigned to Cluster 1 are Declaratives, 16 are Imperatives, 2 are Content Questions, and none are Polar Questions. 8 of the phrases assigned to Cluster 2 are Declaratives, 15 are Imperatives, 6 are Content Questions, and 7 are Polar Questions.

The proportional distribution of Cluster over Sentence Type is displayed in Figure 4 and the exact numbers of each Cluster in each Sentence Type is displayed in Table 2. The distribution of the two Clusters over the Sentence Types helps to illuminate some of the relationships between intonation and sentence type in Ende. First, there is a strong association of interrogative Sentence Types with Cluster 2: 75% of the contours from Content Questions 100% of the Polar Questions are sorted into Cluster 2. Second, the Declaratives are more strongly associated with Cluster 1 (71%) than Cluster 2 (29%). Imperatives, on the other hand, are rather evenly split with 52% falling into Cluster 1 and 48% falling into Cluster 2.

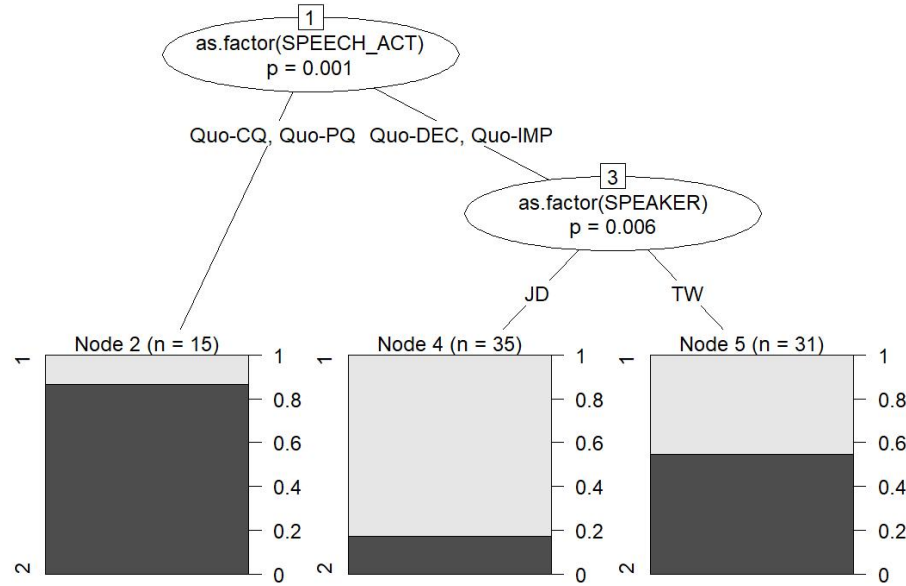


Figure 5: Conditional inference tree with Cluster as the dependent variable and with Sentence Type, Speaker, Duration of interval (in seconds), and Read vs. Retelling as the independent variables. The first split is according to Sentence Type; Content Questions (Quo-CQ) and Polar Questions (Quo-PQ) form one branch while Declaratives (DEC) and Imperatives (IMP) form the second branch. There is no additional split within the interrogative Sentence Types with over 80% of these phrases being in Cluster 2. The non-interrogative Sentence Types are further split according to Speaker. Over 80% of the non-interrogative phrases uttered by JD are in Cluster 1. Between 40 and 50% of the phrases uttered by TW are in Cluster 1.

Figure 5 shows a conditional inference tree with Cluster as the dependent variable and with Sentence Type, Speaker, Duration of interval (in seconds), and Read vs. Retelling as the dependent variable. This indicates that the strongest predictor of Cluster is Sentence Type ($p = .001$) which splits the dataset according to interrogative vs. non-interrogative phrases. A contour is much more likely to be in Cluster 2 if its phrase has a Sentence Type of Content Question (Quo-CQ) and Polar Questions (Quo-PQ). The phrases with non-interrogative Sentence Types are further split according to Speaker identity ($p = .006$). Declarative (Quo-DEC) and Imperative (Quo-IMP) are much more likely to be sorted into Cluster 1 than Cluster 2 if they were spoken by Jerry Dareda (JD) while phrases uttered by Tony Warama (TW) are only slightly more likely likely to be sorted into Cluster 2 than they are Cluster 1. The relationship between Speaker and Cluster requires further investigation. It could arise as a consequence of an asymmetry in the non-interrogative Sentence Types used by both speakers. For example, if JD's

narratives contained more Declaratives than Imperatives it might lead naturally to more Cluster 1 contours in JD's speech. The Speaker Min - Max normalization is intended to neutralize differences in the pitch ranges of the speakers while maintaining differences in the pitch ranges within speakers. As such, this may identify a tendency for TW to use a higher and wider pitch range when speaking from the perspective of one of his characters, even when using non-interrogative phrases. JD, on the other hand may not alter his pitch range for non-interrogatives these same contexts.

The conditional inference tree in Figure 5 suggests that Imperatives and Declaratives pattern similarly in contrast to the interrogative sentence types. However, looking at the proportional distribution of the two Clusters within the Imperatives shown in Figure 4, it is clear that both Clusters are much more evenly distributed over the Imperatives than either the interrogative Sentence Types or the Declaratives. This suggests a more gradient relationship between intonation and Sentence Type than a categorical one.

Figure 6 shows the output of a cluster analysis with a 5 cluster solution. Since the clustering algorithm employed by Kaland's (2023) toolkit builds clusters hierarchically, those identified in the 5 cluster split are subsets of the two clusters shown in Figure 1. As such, they may be directly related to one another. In this split, Clusters 1 and 2 are subsets of Cluster 1 in Figure 1 and Clusters 3, 4, and 5 are subsets of Cluster 2 in Figure 1. Figure 7 shows the proportional distribution of each Sentence Type over each Cluster and Figure 8 shows the conditional inference tree with Sentence Type as the dependent variable and with Cluster, Speaker, Duration of interval (in seconds) and Read vs. Retelling.

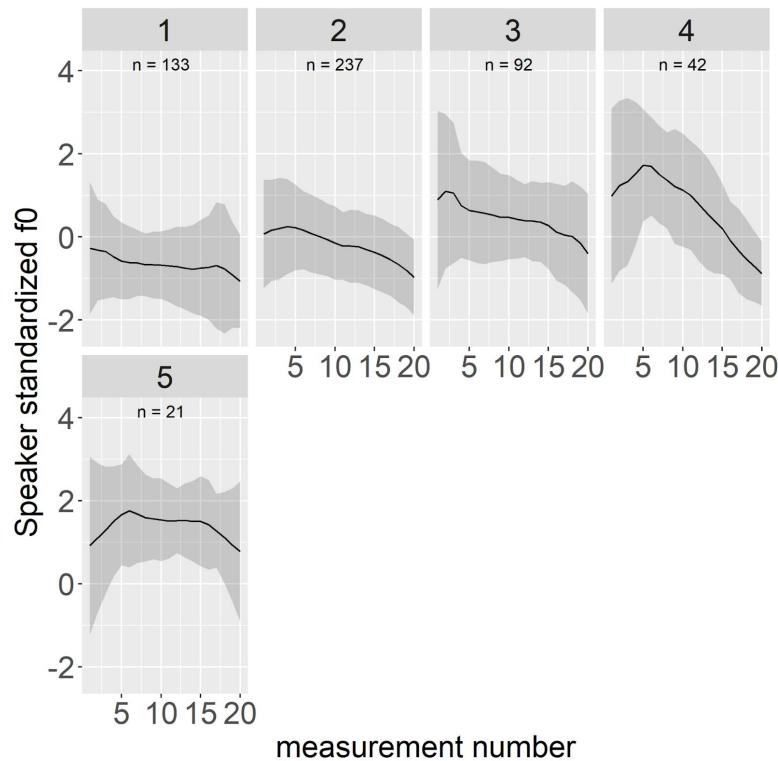


Figure 6: Averaged f_0 contours for a clustering solution with 5 clusters.

The averaged contours for the five cluster split in Figure 6 reveals some of the phonetic variation that was obscured by the larger groupings in Figure 1. However, Figure 7 shows that the distribution of Sentence Type over each Cluster becomes more diffuse as the number of clusters increases.

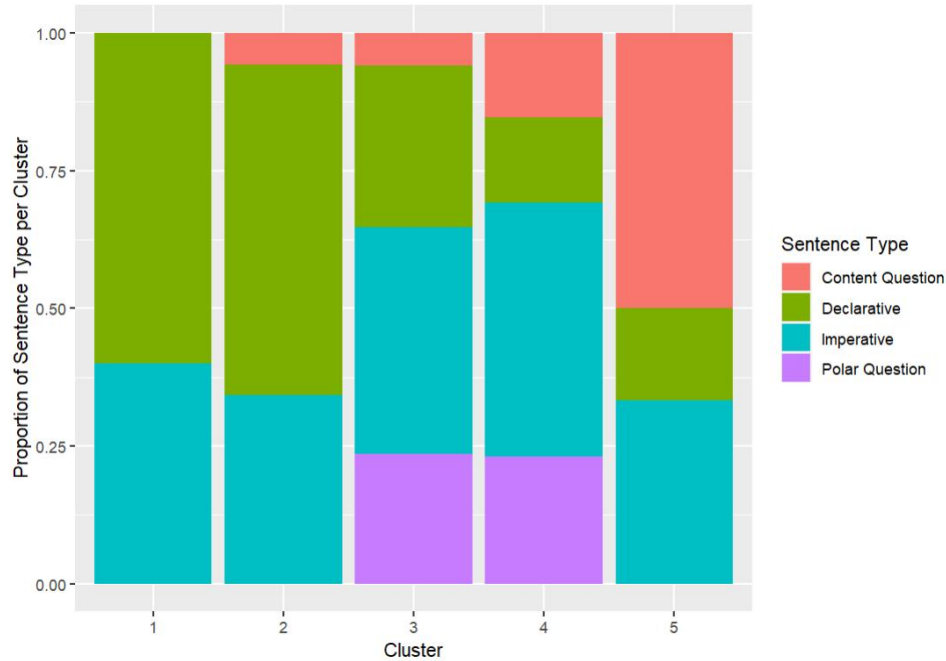


Figure 7: Proportional distribution of Sentence Type over Cluster for the 5 cluster split.

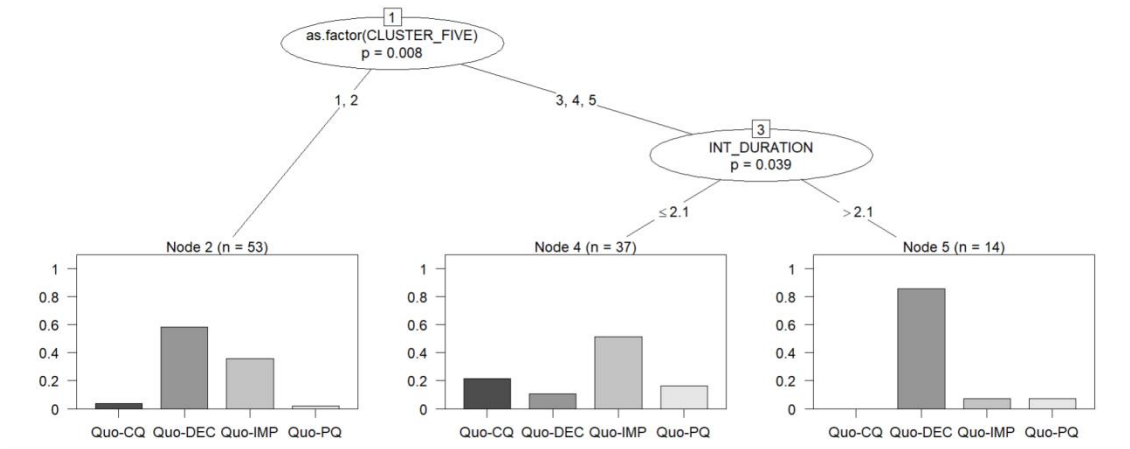


Figure 8: Conditional inference tree for a five cluster solution. Sentence Type is the dependent variable and Cluster, Speaker, Duration (in seconds), and Read vs. Retelling are the independent variables.

The conditional inference tree in Figure 9 shows that the strongest predictor is Cluster ($p = .008$) with a split between Cluster 1 and 2 on the one hand and a Clusters 3, 4 and 5 on the other. The phrases with contours sorted into Clusters 1 and 2 are more likely to be Declaratives or Imperatives than they are to be Content Questions or Polar Questions. The majority of the

Content and Polar Questions are sorted into Cluster 3, 4, and 5. Duration (in seconds) is a secondary predictor ($p = .039$) with most of the Declaratives with contours in these three clusters being longer than 2.1 seconds in duration. However, the fact that Clusters 1 and 2 are subsets of the original Cluster 1 in Figure 1 and Clusters 3, 4, and 5 are subsets of the original Cluster 2 suggests that increasing the number of clusters does not improve our understanding of the relationship between intonation and Sentence Type in Ende.

Although increasing the number of clusters does not appear to shed more light on the relationship between intonation and sentence type in Ende, it does give us more phonetic detail concerning the variation among intonational contours in Ende. The bulk of the variation shown in the five cluster split (Figure 6) shows variation according to overall pitch height, the alignment of a pitch peak, and the slope and timing of the pitch fall to the end of the phrase. Notably, none of the identified clusters show averaged contours with a medial or final pitch rise. This seems to indicate a typical intonational contour shape characterized by a rise to the pitch maximum at the beginning of the phrase in Ende followed by a fall to the pitch minimum at the end of the phrase.

4.1.1 Interim discussion

The cluster analysis presented above is unable to identify a clear mapping of f0 contours onto different Sentence Types like *declarative*, *interrogative*, or *imperative*. Additionally, the averaged contours calculated for each cluster at both the two cluster and the five cluster splits indicate that the typical intonational contour of an Ende phrase is characterized by a pitch peak to the global maximum near the beginning of the phrase and a pitch fall to the global minimum near the end of the phrase. This is in line with previous descriptions of Ende's intonation which state that there is not a systematic difference between *declarative* and *interrogative* phrases in Ende and that both are often associated with falling intonational contours (Lindsey 2022). This analysis additionally suggests that the intonational contours associated *imperative* phrases may share this description.

As it stands, it does not appear that intonation is playing the primary role in distinguishing between Sentence Type in the Ende phrases contained in this dataset. However, there is still an apparent relationship between Sentence Type and intonation in Ende. The distributional analyses of the two cluster split reveal that the *interrogative* phrases are more closely associated with a pitch contour exhibiting a higher and a wider average range while the *declarative* phrases are more closely associated with a pitch contour with a lower and more narrow average contour. These correlations may be related to cross-linguistic tendencies for high/rising pitch ranges to be employed in order to communicate attitudes of politeness, lack of confidence or deference, while low/falling pitch contours to communicate attitudes of assertion, aggression, or dominance (Bolinger 1978; Ohala 1984, 1996). The fact that the *imperative* phrases are fairly equally distributed across the two clusters may further hint at this relationship, a command given in excitement would be expected to have a higher pitch while a command given from a position of dominance might be expected to have a lower f0.

Additionally, the conditional inference tree in Figure 3 indicates that phrases in Cluster 2 are more likely to be Declarative if they are longer in duration. Phrases with longer duration may have higher pitch at the beginning of the phrase in order to support a given rate of decline over the utterance (Cooper & Sorenson 1981 as cited in Ladd 1984). Since the f0 measurement

procedure employed by Kaland's (2023) toolkit converts each intonational contour to the measurements taken at 20 evenly spaced points along the contour, the contours over longer phrases are compressed and might appear to have more dynamic intonational profiles. If this is the case, the longer Declaratives in Cluster 2 may have more similar profiles to the averaged contour shown for Cluster 1. This would serve to strengthen the relationships between affect and intonation outlined above. Confirming and understanding these potential relationships between intonation speaker affect, attitude, and stance in Ende will require further investigation.

Although the two cluster split appears to be the most informative for understanding the relationship between intonation and sentence type in Ende, the phonetic variation identified by the averaged contours at splits with higher numbers of clusters may offer promising avenues for future research. For instance, the differences in pitch height at the beginning of a phrase may be related to different aspects of narrative structure. However, the fact that the averaged contours presented for the five cluster split (Figure 6) are all characterized by a rise in pitch to the pitch maximum near the beginning of the phrase and a fall in pitch near the end of the phrase may be telling. It may be the case that contours with non-falling pitch movements constitute a small subset and are obscured by the larger clusters. However, previous applications of this clustering algorithm suggest that contours with medial and final pitch rises are fairly easily differentiated from those with overall falling pitch and often constitute clusters that are identified early in the clustering split (Babinski 2022; Babinski & Bower 2022; Kaland 2023). This suggests that the intonational contour shape characterized by a pitch peak near the beginning of the phrase and a pitch fall to the end of the phrase may be typical across Ende phrases. As such, it appears that the primary role that intonation is playing in Ende is in delineating the boundaries of prosodic units.

In summation, the results of the cluster analysis leave us with two major generalizations (1) intonation is not used to distinguish between sentence types in Ende and (2) the Ende phrase is delimited by a pitch peak at the beginning of the phrase and a fall at the end of the phrase. In Section 4.2 below, these two generalizations will be treated as predictions are tested with an analysis of the distribution of pitch maxima and pitch minima in the phrases of two independent sound files (see Section 3.1.2) that were not included in the cluster analysis.

4.2 Qualitative analysis

The cluster analysis in Section 4.1 presented two different conclusions - (1) intonation does not play a straightforward role distinguishing between *declarative*, *interrogative*, or *imperative* sentence types in Ende and (2) Ende phrases seem to be delimited by a pitch peak at the beginning of the phrase and a fall at the end of the phrase. In the discussion that follows, these two conclusions will be treated as hypotheses that will be tested through a qualitative analysis of the phrases in two independent files that were not included in the cluster analysis. The first file is a dictionary elicitation in which the speaker, Tony Warama, reads a word in isolation, the definition, and then an example sentence (Warama 2017). This file was selected because it was easy to identify the beginnings and endings phrases. Each of the example sentences constitute one phrase. However, the format of this elicitation may have led the speaker to produce phrases with a consistent intonation contour which otherwise would not have been used (e.g. list intonation). The second file is a "functions of language" elicitation with two speakers Linda Duiya and Anna Duiya (Duiya & Duiya 2018). Here, the two speakers are recorded having

everyday interactions in Ende. Each interaction is prompted by the researcher and there is discussion and coaching between the two speakers as they construct the interaction. Although this may lead to some artificial language use, it does present easily identifiable Ende phrases in a clear discourse.

The procedure for analyzing the dictionary and functions of language elicitations is as follows. Phrases were segmented according to the same criteria used to segment the phrases in the cluster analysis in Section 4.1. Each phrase was inspected (semi-)impressionistically. Each phrase was listened to and inspected with the pitch tracker in Praat (Boersma & Weenink 2018). First the highest point along the pitch track (the pitch maximum) within each phrase is marked with an H and the lowest point along the pitch track (the pitch minimum) within each phrase is marked with an L. Although these symbols are borrowed from the ToBI annotation system they are merely intended to indicate the pitch minima, the pitch maxima the phonetic signal, not underlying phonological categories. In the following subsections, only annotations that are relevant to the discussion are included. All generalizations that are drawn in this section will require further research to confirm.

4.3 Intonation and sentence type

The first prediction is that intonation is not used to distinguish sentence types in Ende. The elicitations used for this investigation were not intended directly address differences between sentence types. Because of this, I have not been able to find directly comparable sentences like ‘Tony built a new canoe.’ vs. ‘Did Tony build a new canoe?’. As a consequence, this prediction will be addressed by looking at examples of *declaratives*, *content questions*, *polar questions*, and *imperatives* and showing that their melodies follow the same general shape and can be broken down into essentially the same component parts. Namely, a pitch peak within the first word of the phrase and a pitch fall which terminates at the end of the phrase. The following examples demonstrate this pattern with a declarative sentence (example 1), a content question (example 2), a polar question (example 3), and an imperative (example 4). In all cases, the global pitch maximum is marked with an H and the global pitch minimum is marked with an L.

First, looking at Declarative phrases, we have *Boge kollba da mer abal ddäddäg dan* ‘Mudfish is a good animal’. Figure 9 shows the spectrogram and pitch track for the phrase while the gloss and translation is provided in example 1. The pitch maximum is marked with an H and the pitch minimum is marked with an L.

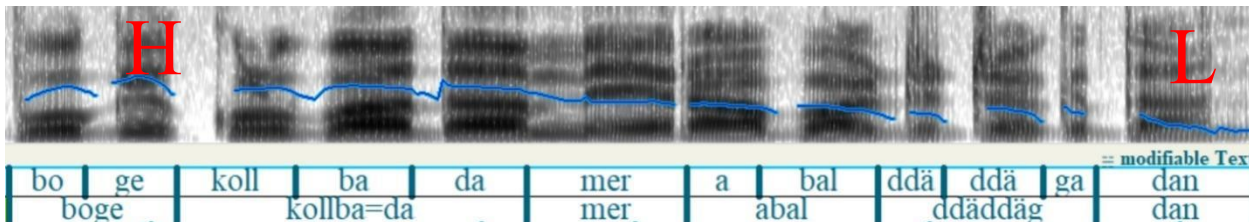


Figure 9: Spectrogram and pitch track for example 1.

- (1) Boge kollba da mer abal ddäddäg dan
 Boge kollba=da mer abal ddäddäg dan
 Mudfish fish=NOM good very animal cop

“Mudfish is very delicious.”

Warama 2017: 62.28 – 64.88

Here we see that pitch maximum is realized somewhere within the second syllable of the first word in the phrase *boge* ‘mudfish’. The pitch minimum, on the other hand, is realized in the last syllable of the phrase which also corresponds with the final word of the phrase, the copula *dan*.

Second, looking at Content Questions, we have the example *Amo gall sisor dan dābe* ‘whose new canoe is this?’. Figure 10 shows the spectrogram and pitch track for the phrase while the gloss and translation is provided in example 2. The pitch maximum is marked with an H and the pitch minimum is marked with an L.

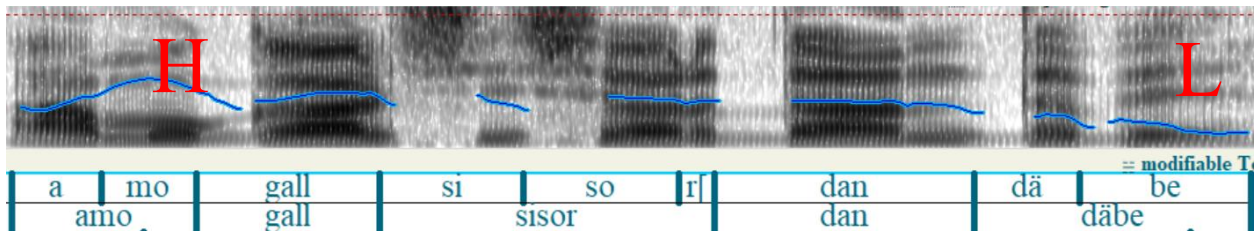


Figure 10: Spectrogram and pitch track for example 2.

(2) Amo gall sisor dan dābe?

Amo gall sisor dan dābe

Who.sg.poss canoe new cop this

“Whose new canoe is this?”

Warama 2017: 94.84 – 96.54

Again, we see that the pitch maximum is realized within the second syllable of the first word in the phrase *amo* ‘who (sg. Possessive)’. The pitch minimum corresponds with the final syllable of the phrase, in this case that is the second syllable of *dābe* ‘this’.

Third, looking at Polar Questions, we have the example *Bongo bāne igi pite de zāme amāttalle* ‘Are you wearing underpants? (lit. have you already put on your underwear?)’ Figure 11 shows the spectrogram and pitch track for the phrase while the gloss and translation is provided in example 3. The pitch maximum is marked with an H and the pitch minimum is marked with an L.

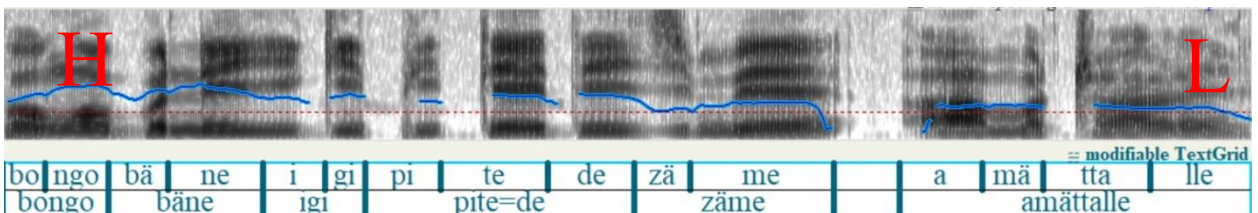


Figure 11: Spectrogram and pitch track for the sentence in example 3.

- (3) Bongo bāne igi pite de zāme amāttalle
 Bongo bāne igi pite=de zāme amāttalle
 2.sg.nom 2.sg.poss underwear=ACC already put.on

“Are you wearing underpants? (lit. have you already put on your underwear.)”

Warama 2017: 328.29 – 331.33

Once again, the pitch maximum occurs in the second syllable of the first word *bongo* ‘you (sg. nom.)’ and the pitch minimum corresponds with the last syllable of the phrase, in this case the final syllable of the verb *amāttalle* ‘put on’.

Finally, looking at Imperatives, we have the example *Bāne turik de abo iweny gall tit i* ‘Bring your axe to dig the canoe!’ Figure 12 shows the spectrogram and pitch track for the phrase while the gloss and translation is provided in example 4. The pitch maximum is marked with an H and the pitch minimum is marked with an L.

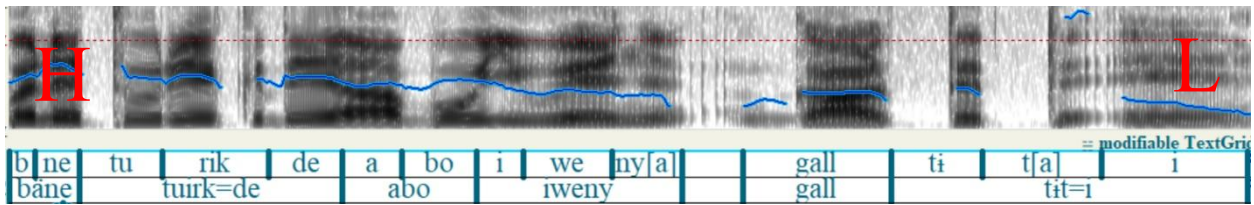


Figure 12: Spectrogram and pitch track for the sentence in example 4.

- (4) Bāne turik de abo iweny gall tit i
 Bāne turik=de abo iweny gall tit=i
 2.sg.poss axe=ACC must carry canoe hollow=ALL

“Bring your axe to dig the canoe!”

Warama 2017: 136.26 – 139.49

Here we see an imperative with the same general f0 contour as the declarative, content question, and polar question. The pitch maximum again falls in the second syllable of the first word *bāne* ‘2nd person singular possessive’ and the pitch minimum is reached on the last syllable of the phrase, in this case the allative enclitic *i* on the verb *tit* ‘hollow’.

The phrases representing each of the sentence types tested in this investigation show globally falling f0 contours that are characterized by a pitch peak within the first word and a pitch fall terminating in the last syllable of the phrase. This shape is in line with the averaged f0 contours identified in the cluster analysis. As such, there is no evidence, either from the distribution of sentence types across identified contour clusters, global contour shape, or differences in local pitch movements associated with the edges of phrases to suggest that intonation plays a role in distinguishing sentence type in Ende.

5 Discussion

5.1 Intonation and sentence type

The results of the cluster analysis and the inspection of f0 contours in the dictionary and ‘functions of language’ elicitations suggest that the typical Ende phrase is characterized by a rise in pitch to the pitch maximum within the first word of the phrase and a global fall to the pitch minimum which is reached in the last syllable of the phrase. This corroborates previous descriptions of Ende’s intonational system (Lindsey 2022) and extends the description of a generally falling contour to *imperative* phrases. While intonation does not appear to play a straight forward role in distinguishing between sentence types in Ende, there does appear to be a relationship between interrogative phrases and a heightened, widened pitch range. For a more detailed discussion of this relationship and the interpretation of the cluster analysis which provides the principle evidence towards it, I refer the reader to Section 4.1.1.

The globally falling pattern identified by the averaged contours may be related to the phenomena of declination (Ladd 1984). Declination describes the cross-linguistic tendency for the pitch range to lower and narrow over the course of an utterance. While this is likely a factor in the observed pattern, the fact that phrases inspected in this investigation systematically end with a fall in pitch may suggest the presence of something like a low boundary tone marking the end of the phrase in Ende.

The observation that intonation does not seem to play a role in distinguishing between sentence types in Ende fits with the description of Coastal Marind (Olsson 2018), Yelmek (Gregor 2021) and Nmbo (Kashimi 2021). Since Coastal Marind (Anim), Yelmek (Yelmek-Maklew family), Nmbo (Yam family)⁷, and Ende (Pahoturi River family) are unrelated and spoken in geographically disparate areas, it is possible that this is a feature of Southern New Guinea. This, however, is complicated by the descriptions of rising intonation as a signal to interrogative phrases in Komnzo (Döhler 2018), Meryim Mir (Piper 1989), and Nama (Siegel 2023).

While the cluster analysis presented in Section 4.1 identified the general shape of intonational contours in Ende to be characterized as a generally falling with a rise to the pitch maximum at the beginning of the phrase and a fall to the pitch minimum at the end of the phrase, the distributions of pitch maxima and pitch minima outlined in Section 4.2 show that this pattern presents itself consistently across phrases from contexts outside of the narratives used for the cluster analysis. This suggests that intonation plays a role in delimiting the boundaries of prosodic units in Ende, with the a rise to the pitch maximum occurring within the first word of the phrase and the pitch minimum being realized in the final syllable of the phrase. Furthermore, these movements are consistent across phrases of various sentence types.

5.2 Intonation as a cue to prosodic units

Thus far, this investigation has identified a consistent contour shape that is associated with the Ende phrase. This contour is characterized by a pitch peak at the beginning of the phrase and a

⁷ Gregor (2021) reports the presence of non-low boundary tones in Nmbo but does not speculate as to their function.

pitch fall at the end of the phrase. As such, it appears that intonation plays a role in delineating phrases in Ende. The remainder of this section discusses observations concerning the timing of both the pitch rise at the beginning of the word and the pitch fall at the end of the word.

Without the intuition and aid of native speakers, it is difficult to determine what pitch movements may be meaningful in Ende. As such, the annotations that are provided in this section are only intended to indicate phonetic movements in pitch; they are not intended as claims concerning phonological categories or the exact nature of the alignment of these movements within the Ende word and phrase. The annotations were made in two rounds. First, the pitch maximum for each phrase was marked with an H and the pitch minimum for each phrase was marked with an L. Then, f0 measurements (in semitones) were taken at the temporal midpoint of each syllable within the phrase. A threshold of difference exceeding 1.2 semitones from one syllable to the next was used to determine whether or not a syllable would be annotated for a potential pitch movement. The primary motivation for this threshold was to provide a criteria for annotating potentially meaningful pitch movements in a consistent and replicable way that was not dependent on the judgement of the investigator (an monolingual English speaking linguist). The exact threshold of 1.2 semitones was decided upon because a threshold of 1 semitone seemed to overproduce labels while a threshold of 1.5 semitones seemed to many seemingly significant pitch movements. Future iterations of this investigation would benefit from a more principled method of identifying potentially meaningful pitch movements that does not rely on the application of an arbitrary threshold. In any instance where the pitch increased by an amount of 1.2 semitones or greater from the midpoint of one syllable to the next an H was used to mark the midpoint of the syllable in which that increase was achieved. Similarly, any instance where the pitch decreased by an amount of 1.2 semitones or greater from the midpoint of one syllable to the next and L was used to mark the midpoint of the syllable in which that decrease was achieved. The relationship between pitch measurements at consecutive midpoints will be referred to as a “local pitch movement” in the discussion which follows. In cases where this procedure would redundantly mark syllable already annotated with an H or L for the pitch maximum or pitch minimum, no additional annotation was made.

5.2.1 Pitch peaks at the left edge of the phrase

The cases outlined above in Section 4.2 demonstrate that the pitch maximum of the phrase in Ende is reached within the first word of the phrase. In all of those cases the pitch maximum was reached in the second syllable of the word. Each of the phrases discussed in examples 1 through 4 started with a two syllable word. In the figures 13 through 16, an H indicates the pitch maximum and an L indicates a local fall from the midpoint of the proceeding syllable to the midpoint of the annotated syllable.

However, in cases where the first word in the phrase is only one syllable, the placement of the pitch maximum is reached within that syllable and is followed by a fall in pitch on the first syllable of the following word. Figure 13 shows the spectrogram and pitch track of the first two words *ge baba* ‘this father’ in the phrase *ge baba bāne digol dompa popoatt dan*. ‘Father made this dompa out of digol tree.’ which is glossed and translated in example 5.

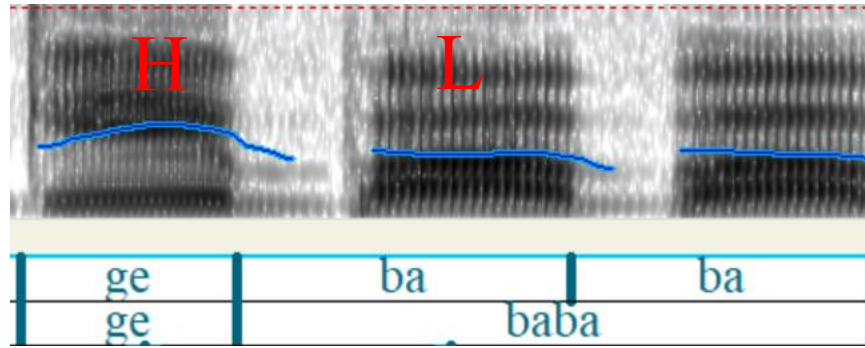


Figure 13: Spectrogram and pitch track of the first two words in example 5.

- (5) Ge baba bāne digol dompa popo=att dan.
 Ge baba=bāne digol dompa popo=att dan.
 This father=3.sg.pst.poss tree arrow sharpen=ABL cop
 “Father made this dompa out of digol tree.”
 Warama 2017: 298.54– 301.85

In this case, the pitch maximum is contained within the single syllable of the first word *ge* ‘this’. This is followed by a step down in pitch on the first syllable of the following word *baba*.

Next, when the first word of the phrase is longer than two syllables long, the pitch maximum is still reached in the second syllable. However, the pitch remains relatively high throughout the remainder of that word. Figure 14 shows the spectrogram and pitch track of the first two words *masamasar tāräp* ‘old times’ in the sentence *Masamasar tāräp me ngattong kanas toboll de dokomneyo* ‘In the old times, one man carried the kanas arrows.’ which is glossed and translated in example 6.

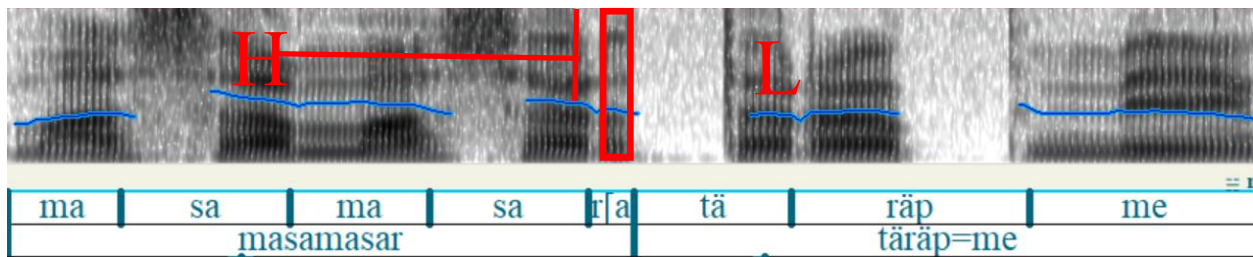


Figure 14: Spectrogram and pitch track for the sentence in example 6.

- (6) Masamasar tāräp me ngattong kanas toboll de dokomneyo
 Masamasar tāräp=me ngattong kanas toboll=de dokomneyo
 Ancestors time=LOC first arrow spear=ACC carried
 “In the old times, one man carried the kanas arrows.”

Warama 2017: 270.92 – 275.21

Here even though the first word of the phrase has four syllables, the pitch maximum is reached in the second syllable of *masamasar* ‘ancestors’. The pitch remains relatively high

through the duration of the word, as indicated by the red bar, before stepping down on the first syllable of the second word *täräp* ‘time’. Note also that there is an epenthetic vowel, between *masamasar* and *täräp*. This is outlined in red in Figure 13. Although the pitch fall does not cross the threshold of 1.2 semitones until the first syllable of *täräp*, the fall appears to begin during the epenthetic vowel.

There are also cases where the first word in the phrase is produced with an epenthetic vowel. Figure 15 shows the first word *mänmän* ‘girls’ of the sentence *Mänmän a sana de dändäraebeyo zazaba we* ‘The girls filled their sago bags with sago.’ glossed and translated in example 7.

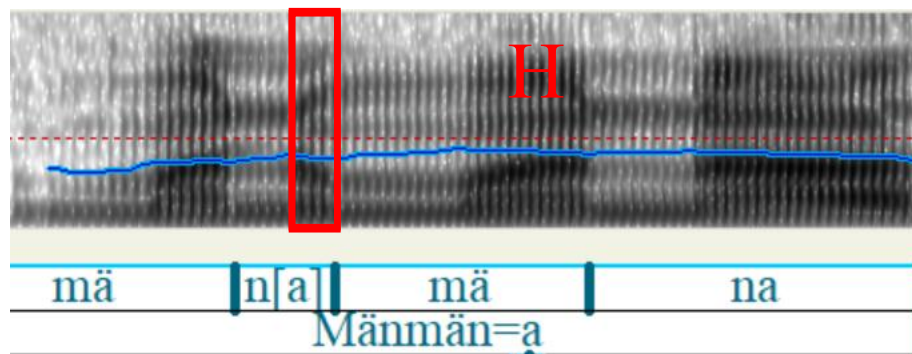


Figure 15: Spectrogram and pitch track for the first word of example 7.

- (7) Mänmän a sana de dändäraebeyo zazaba we
 Mänmän=a sana=de dändäraebeyo zazaba=we
 Girls=NOM sago=ACC filled bag=ALL

“The girls filled their sago bags with sago.”

Warama 2017: 209.73 – 214.36

Here, *mänmän* ‘girls’ is pronounced with an epenthetic vowel separating the first and second underlying syllables of the word. The epenthetic vowel is outlined with a red box. Although the epenthetic vowel is phonetically the second syllable of the phrase, the pitch maximum is not reached until the second underlying syllable of the phrase. This suggests that the alignment of the pitch rise at the beginning of the phrase may target the second underlying syllable. However, this will need further investigation.

There are also cases where the first word in the phrase is followed by a nominal enclitic. In these cases, the pitch peak still occurs on the second syllable of the first word and the pitch remains relatively high through the duration of the enclitic, just as it would through the duration of a multisyllabic word as we see with *masamasar* ‘ancestors’ in Figure 14 above. Figure 16 shows the first two words *mälla* ‘woman’ of the phrase *Mälla da kemibi kollba de nägäddallo gull alle* ‘The women caught many fish with the net.’ which is glossed and translated in example 8.

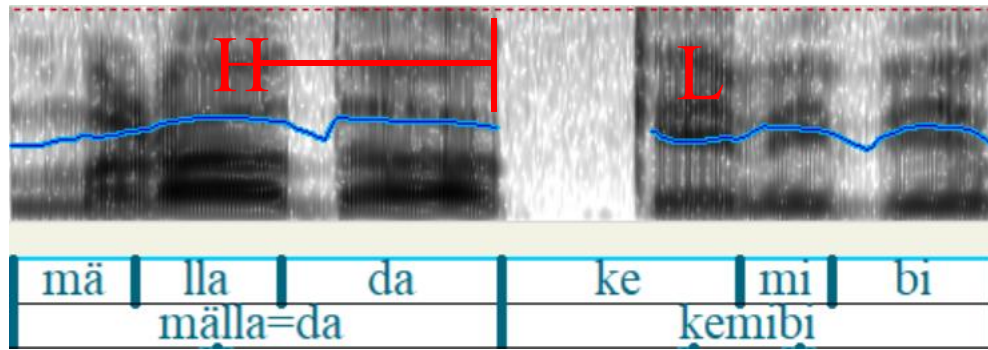


Figure 16: Spectrogram and pitch track for the first two words in example 8.

- (8) Mälla da kemibi kollba de năgăddallo gull alle.
 Mälla=da kemibi kollba=de năgăddallo gull=alle.
 Woman=NOM many fish=ACC kill net=INS
 “The women caught many fish with the net.”
 Warama 2017: 247.8 - 252.87

Here the pitch maximum is in the second syllable of *mälla* ‘woman’, however, the pitch remains high through the duration of the nominative enclitic *da* before stepping down on the first syllable of the following word *kemibi* ‘many’. This may serve as evidence that the nominal enclitic forms a prosodic word with the word immediately preceding it.

To summarize, the pitch maximum in the Ende phrase tends to be realized in the second underlying syllable of the first word of a phrase in Ende. When the first word of the phrase is monosyllabic, the pitch maximum is realized in that single syllable. This suggests that the alignment is sensitive to word hood and is not simply aligning with the second syllable of the phrase. Additionally, the fact that the pitch remains high through the duration of words longer than two syllables, including enclitics suggests that the intonational system is sensitive to word boundaries.

5.2.2 Pitch fall at the right edge of the phrase

As we saw in the examples from Section 4.2 and in the averaged f0 contours from the cluster analysis, the typical Ende phrase has a fall to the pitch minimum marking the end of the phrase. As with the patterns observed in the pitch maximum associated with the first word of the phrase described above, this pitch fall seems sensitive to word hood. This section describes the patterns of pitch falls associated with the right edge of the Ende phrase. In the figures that follow, local pitch falls of more than 1.2 semitones are marked with an L. In all cases the global pitch minimum is reached on the last syllable of the phrase. The generalizations concerning the pitch falls reported in this section are more speculative than those concerning the pitch peaks reported in Section 5.2.1. This is pitch peaks associated with the first word in the phrases constitute a clear turning point in the pitch track while these local falls mostly appear as steepening slopes along a gradual f0 decline. While there are potential pitch falls throughout the duration of the phrase, these have not yet been analyzed and may not straightforwardly be related to word

boundaries. Again, these annotations are not intended to indicate claims about underlying phonological categories.

Figure 17 shows the spectrogram and pitch track for the final word *dae* and the syllable immediately preceding the final word *dae deyakonewän* ‘along covered’ in the phrase *däbe ako ddapall ge enanae dae deyakonewän* ‘On that day, the sky was covered.’ which is glossed and translated in example 9. Here we see a pitch fall on the first and second syllables of the final word *deyakonewän* ‘covered’. The pitch remains relatively level until the final syllable where the pitch falls again to its minimum.

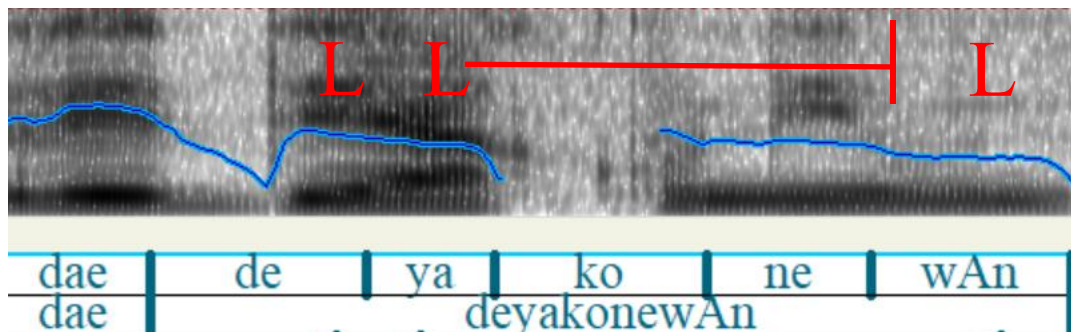


Figure 17: Spectrogram and pitch track for the last two words in example 9.

- (9) *däbe ako ddapall ge enanae dae deyakonewän.*
däbe ako ddapall ge enanae=dae deyakonewän
 that then sky this straight=PEARL covered
 “On that day, the sky was covered.”
 Duiya & Duiya 2018: 412.6 – 414.9

When the last word is a copula, there may be a step down on the first syllable of the word immediately preceding the copula. Figure 18 shows the pitch track for the last three words *abal* ‘very’, *ddäddäg* ‘animal’, and *dan* ‘is’ from the phrase *Boge kollba da mer abal ddäddäg dan* ‘Mudfish is a very good animal.’ glossed and translated in example 10. Although the last word of the phrase is the copula *dan*, there is an initial pitch fall on the first syllable of the word preceding the copula *ddäddäg* ‘animal’, the pitch then remains relatively level until it falls to the pitch minimum in the copula. If the initial pitch fall is associated with the first syllable of the final word, this may be evidence that the copula is cliticizing to the preceding word to form a single prosodic word.

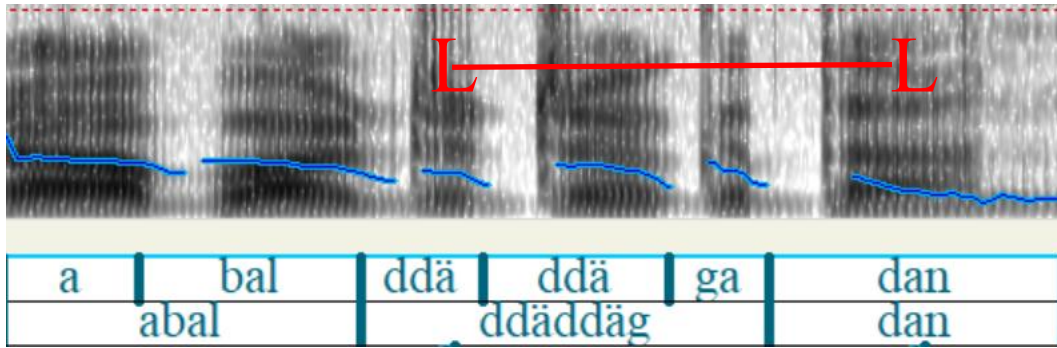


Figure 18: Spectrogram and pitch track for the copula and the preceding two words from example 10.

- (10) Boge kollba da mer abal ddäddäg dan
 Boge kollba=da mer abal ddäddäg dan
 Mudfish fish=NOM good very animal cop
 “Mudfish is a very good fish.”
 Warama 2017: 62.29 – 64.88

A similar pattern is seen when the final word of the phrase is an auxiliary. Figure 19 shows the spectrogram and pitch track for the last three words ‘ause’ old, *abal* ‘very’, and *allan* ‘has’ of the phrase *Kak säre zäme ause abal allan* ‘Grandmother has sadly already grown very old’ which is glossed and translated in example 11. Here the final word of the phrase is the auxiliary *allan*. However, there is a fall in pitch coinciding with the first syllable of the preceding word *abal* ‘very’. The pitch then falls through the duration of the *allan*. Again, if the initial step down in pitch is associated with the first syllable of the final word, then this may be evidence that the auxiliary *allan* is cliticized to the preceding word.

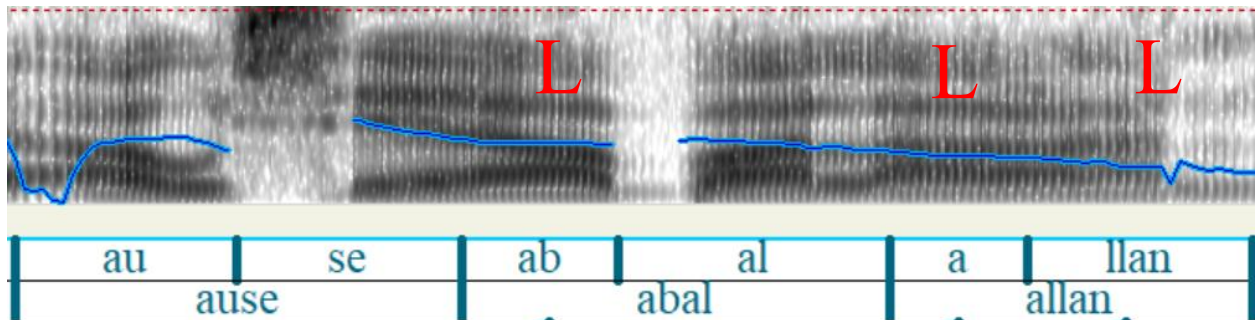


Figure 19: Spectrogram and pitch track of the last three words from example 11.

- (11) Kak säre zäme ause abal allan
 Kak säre zäme ause abal allan
 Grandmother sadly already old very aux
 “Grandmother has sadly already growing very old.”
 Warama 2017: 619.02 – 622.08

There tends to be a fall in pitch coinciding with the first syllable of the final word in a phrase, the pitch minimum is then met on the final syllable of the phrase. As with the behavior of the

pitch peak at the beginning of the phrase, this pattern may provide evidence for wordhood in Ende. However, this claim is speculative and further evidence to support it.

In addition to the fall in pitch, there is often a change from modal to breathy voicing or devoicing during the final word of the phrase. However, this has not been fully explored so the exact character of the change in phonation and its timing relative to the final pitch movements is not well understood.

5.2.3 Evidence for prosodic units between the word and the phrase level

Above we have established that the phrase in Ende appears to be delimited by a pitch rise to the pitch maximum which is realized within the first word in the phrase and a pitch fall which terminates on the final syllable of the phrase. However, there are cases where there may be multiple pitch peaks before the fall at the end of the phrase. In these cases, it appears that the secondary pitch peaks are signaling the beginning of an embedded clause.

Figure 20 shows the spectrogram and the pitch track for the phrase *Ngäna ma we yagnen anggan ngäna era ma me bodmen* ‘I am looking for a house where I can stay in the house.’ which is glossed and translated in example 11. Here there is a clear initial pitch peak in second the first word in the phrase *ngäna* ‘1st person singular nominative pronoun’. The pitch then steps down into the middle of the pitch range where it stays until stepping back up on the second syllable of the next instance of *ngäna*. These points are marked with an H in Figure 20.

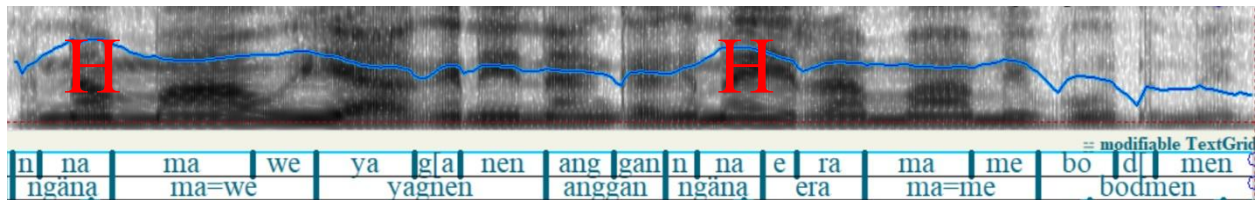


Figure 20: Spectrogram and pitch track for the phrase in example 12.

(12) Ngäna ma we yagnen anggan ngäna era ma me bodmen

ngäna ma=we yag-nen anggan ngäna era ma=me bodmen
 1.sg.nom house=ALL search-pl aux 1.sg.nom cop house=LOC sit
 "I am looking for a house where I will/can stay in the house."
 Duiya & Duiya 2018:

In this case, the second instance of *ngäna* is the first word in the embedded phrase *ngäna era ma me bodmen* ‘where I will stay in the house’. This structure can be seen in 13 below, with bracketing indicating the matrix and the relative clauses.

(13) [Ngäna ma we yagnen anggan [ngäna era ma me bodmen]_{relative}]_{matrix}

The presence of two distinct pitch peaks preceding the final fall to the pitch minimum at the end of the phrase may be evidence that Ende’s prosodic system flattens the recursive structure

sketched in example 13 into a flattened structure sketched below in example 14 (Nespor & Vogel 2007).

- (14) [[Ngäna ma we yagnen anggan]_{smaller} [ngäna era ma me bodmen]_{smaller}]_{larger}
"I am looking for a house where I will/can stay in the house."

Here the relative clause and the matrix clause are both parsed into two prosodic units of the same type within the larger prosodic unit which I have been referring to as the ‘phrase’ or ‘sentence’. The right edge of this prosodic unit is then signaled with a pitch peak while the left edge of the higher prosodic unit is marked with a fall to the pitch minimum. As such, it appears that the patterns of pitch peaks and pitch falls may play a role in signaling hierarchical prosodic structure in Ende, as well as word-hood.

5.3 Limitations

This section outlines a number of limitations which affected the progression and outcomes of the study presented in this paper. First, the difficulties surrounding travel to the intended field site in Papua New Guinea led to a situation in which the investigation was constrained to the existing data on Ende. Although this data was not collected with the intonational research in mind, it provided an opportunity to construct a case study for the application of an intonational cluster analysis workflow (Kaland 2023) to a corpus of spontaneous Ende speech. However, since the phrases used for the cluster analysis were not segmented and annotated by native Ende speakers, the way in which the observations concerning pitch movements as indicators of phrase boundaries will have to be confirmed in the future.

Additionally, the decision to construct the corpus for the cluster analysis from narratives produced by two male speakers in their 30s may limit the generalizability of the findings presented in this study. First, constructing the corpus out of narratives may lead the observed intonational patterns to be an artifact of something like a ‘narrative register’ and not a property of Ende speech in general. Additionally, the use of speech from two male speakers of similar ages in the corpus limits the possibility of identifying patterns which may be aligned with social factors like age and sex/gender. More generally, the use of narratives led to a large asymmetry between Sentence Types in the dataset. As suggested by Babinski & Bower (2023), this asymmetry may obscure intonational similarities that could distinguish these Sentence Types. While some of these concerns may be mitigated by the manual inspection of the auxiliary dataset in Sections 5, future investigations would benefit greatly from a more balanced dataset with data ranging from a more diverse set of speaking contexts with a more diverse range of speakers. Fortunately, the recordings already available in *The Language Corpus of Ende* (Lindsey 2015) provide opportunities to address these limitations.

6 Conclusion

This study addresses the claim that intonation is not used to distinguish between sentence types in Ende, a Pahoturi River language spoken in Southern New Guinea. This is done by applying a cluster analysis workflow developed by Kaland (2023) to a small corpus of Ende narratives and inspecting the f0 contours of two additional independent sound files. The

distribution of sentence types across identified clusters suggests that there is not a straightforward mapping between phonetically different f0 contours and sentence type. Additionally, the average f0 contours for the identified clusters suggest that the typical Ende phrase is characterized by a rise to the pitch maximum near the beginning of the phrase, a globally falling f0 contour and a fall at the end of the phrase. Inspecting the two independent sound files confirms that this pattern remains consistent for phrases of different sentence types even outside of the narratives used for the cluster analysis. This suggests that intonation plays a role in delineating the boundaries of Ende phrases.

An evaluation of the phrases contained in the two independent files suggests that the pitch peaks at the beginning and the pitch fall at the end of the phrase may be sensitive to word boundaries. Additionally, the presence of multiple pitch peaks without an intervening pitch minimum suggests the presence of a prosodic constituent between the level of the word and the phrase. The left edge of the lower prosodic unit appears to be signaled with a pitch peak coinciding with the second syllable of the first word in its domain, while the right edge of the higher prosodic unit is signaled with a fall to the phrase's pitch minimum. Each of these generalizations stand as preliminary statements to guide future work on intonation in Ende.

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